COMICCRAFT

AI Comic Story Creator

5. Implementation

Team Information

Team Member	Role
Joseph Thomas	Team Leader
Sugeeve Raj	Team Member
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Application Entry Point

The application creates a FastAPI instance using the configured application name, mounts /static to the configured static directory, includes the application router, and exposes a health endpoint.

Routes

- GET / – displays the ComicCraft homepage.
- POST /generate – accepts form input and returns the comic preview.
- POST /generate-comic/json – accepts PromptRequest JSON and returns the generated layout and PDF path.
- GET /export-success – displays the PDF export completion page.
- GET /test-image – tests image generation.

Generation Pipeline

- Validate the PromptRequest.
- Generate the five-panel outline.
- Expand the outline into StoryPanel objects.
- Generate an image for every panel.
- Build the ComicLayout.
- Save the comic as a PDF.
- Render the browser preview or return JSON.

Image Generation

DEMO_MODE creates deterministic local PNG illustrations. Real generation can use Hugging Face text-to-image or the optional local Diffusers/Stable Diffusion backend.

Configuration

The documented configuration includes DEMO_MODE, GEMINI_API_KEY, Gemini outline/story model IDs, HF_TOKEN, HF_PROVIDER, HF_IMAGE_MODEL, IMAGE_BACKEND, local diffusion model, image width/height/steps, and storage directories.